FERNANDA QUEIROZ SOTERO CAIO

Aerospace Engineering Student

CONTACT

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Fernanda Sotero Caio

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Brasília, Brazil

SKILLS

CAD
CATIA, Fusion 360, AutoCAD

Finite Element Analysis
Ansys Mechanical, Ansys APDL

Computing
MATLAB, Mathcad, Scilab

Manufacturing
Filament winding, composites, CNC milling, 3D printing (Cura)

Other Tools
LaTeX, Mendeley

LANGUAGES

Portuguese	Native
English	Fluent
Spanish	Intermediate
French	Basic

PROFILE

Bachelor's student in aerospace engineering at the University of Brasília (UnB), expected to graduate by the end of 2027, with coursework in solid mechanics, aerospace structures, composite materials, finite element analysis, fluid mechanics and heat transfer.

Since 2025, I have been an undergraduate researcher at the Aerospace Structures and Materials Laboratory (LEMA), working on filament winding of carbon fibre composites.

EXPERIENCE AND RESEARCH

Undergraduate Researcher in Filament Winding — **2025 - Present**
Aerospace Structures and Materials Laboratory (LEMA) - UnB

Official member since 2026, working on filament winding of carbon fibre composites. Measurement of winding angle, pretension and resin volume, and characterisation of their effect on the microstructure and tensile strength of the tows. CAD modelling of components and mandrels, 3D printing, winding-path generation and structural analysis, using CATIA, Fusion 360, AutoCAD, Ansys, MATLAB and Mathcad. Also worked in partnership with the Telecommunications Laboratory (LabTelecom) to produce components in composite materials.

Undergraduate Research Fellow — **2024 - 2025**
Undergraduate Research Scholarship - UnB

Astrophysics project on compact stars (white dwarfs and neutron stars): fermion degeneracy pressure and the Chandrasekhar mass limit.

PUBLICATIONS

Influence of Filament Winding Parameters on the Microstructure and Tensile Strength of Consolidated and Impregnated Carbon Fiber Tows — **Aug 2026**
Second author, oral presentation, 8th Brazilian Conference on Composite Materials (BCCM8)

A Simplified Analytical Solution for the Machine Path Generation in Filament Winding of Rectangular Tubes — **Aug 2026**
First author, oral presentation, 8th Brazilian Conference on Composite Materials (BCCM8)

TEACHING

Undergraduate teaching assistant at UnB in **Aerospace Structural Mechanics**, **Calculus I**, **Calculus III** and **Physics I**, holding office hours.

EDUCATION

B.Sc. in Aerospace Engineering — **2022 - 2027**
University of Brasília (UnB), Brazil

Expected graduation: end of 2027. Five-year programme.

U.S. High School Diploma — **2016 - 2020**
Mizzou K-12 Online, University of Missouri, USA

An online programme combining liberal arts and STEAM coursework, built around research, discussion and project work on global issues, with an emphasis on critical thinking and leadership. Coursework was assessed through exams, essays and quizzes, and the programme leads to a U.S. high school diploma.

EXTENSION PROJECTS

Team member in university extension projects at the University of Brasília:

- **Filament-wound drone structure** (ongoing): feasibility study of a filament-wound selective-spraying structure for a multirotor drone. Working on the filament winding of the structure and on its structural calculations.
- **Exploring Thermofluids Engineering and Competition Vehicles**: course for university students, two editions.
- **Pre-Calculus**: course held at the Faculty of Science and Technology in Engineering (FCTE/UnB) in the second semester of 2023.
- **School outreach**: participation of FCTE in the Leonardo da Vinci School careers fair, and reception of Federal District school students at the Gama campus.

COURSES

California State University, Long Beach, USA — **2018**
English Immersion Programme

Practising English on campus while living as a university student.

Casa Thomas Jefferson, Brasília — **2008 - 2020**
English Language Course

Twelve years of English studies at a Brazil-U.S. binational centre.

EXTRACURRICULAR

Roller Skating — **2024 - 2026**
Regional Championships

Competes in two to three categories per championship, with multiple gold and silver medals.

Artistic (Synchronized) Swimming — **2019**
São Paulo Opening

Two gold medals, in the quartet and duet events.

- **Model United Nations**: participant in World Health Organization (WHO) committee simulations.
- **Piano**: ongoing lessons, with two recitals per year.